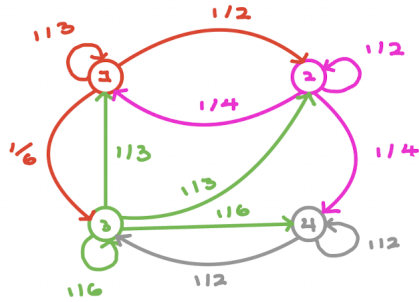


Example 1. A Markov chain $\{X_n\}_{n \geq 0}$ with states $\{1, 2, 3, 4\}$ has transition matrix:

$$P = \begin{bmatrix} \frac{1}{3} & \frac{1}{2} & \frac{1}{6} & 0 \\ \frac{1}{4} & \frac{1}{2} & 0 & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$



Assume the initial state is $X_0 = 1$.

(a) What is the probability that State 3 is reached before State 4?

We define:

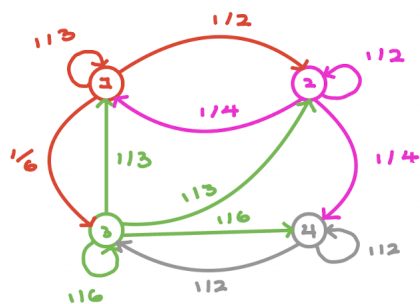
$$p_i =$$

Immediately we know:

$$p_3 = \quad \text{and} \quad p_4 = \quad \text{and our goal is to find:}$$

We use **first step analysis** to devise a system of equations.

Each equation comes from an application of the law of total probability.



Recall: the initial state was 1.

(b) What is expected number of steps until the chain enters either States 3 or 4?

We define:

$$\mu_i =$$

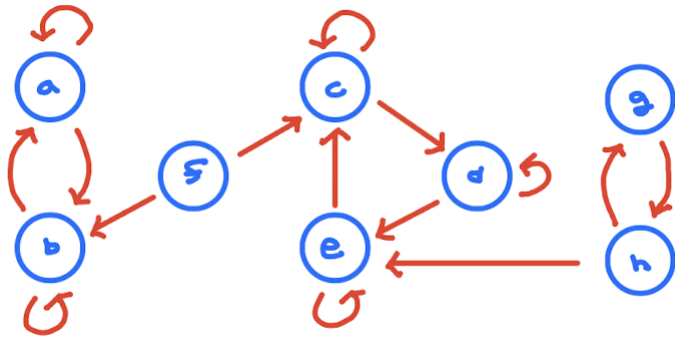
Immediately we know:

$$\mu_3 = \quad \text{and} \quad \mu_4 = \quad \text{and our goal is to find:}$$

We use **first step analysis** to devise a system of equations.

Each equation comes from an application of the law of total expectation, aka the tower property.

A. **Classification of States.** Consider the following Markov diagram, where an arrow indicates a transition that can occur with positive probability.



Because it is possible to eventually transition from state **g** to state **c**, we write:

We say that **c** is **admissible** from **g**.

On the other hand, in the reverse order:

We say that states **a** and **b** **communicate** because:

By definition any state communicates with:

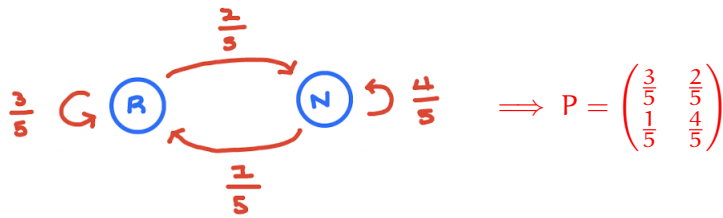
We can thus divide states into **communication classes**:

These are equivalence classes, where the equivalence relation is communication.

A communication class is **transient** if it is possible to transition out of the class and never come back. Otherwise, it is **recurrent**.

If we allow the chain to run indefinitely, it can be shown that a transient class will only be visited finitely many times, while a recurrent class will be visited infinitely often.s

B. **Ergodicity**. Let's return to the Markov chain $\{X_n\}_{n \geq 0}$ of rainy, **R**, versus not-rainy, **N**, days.



Q. Over the long-term, what is the proportion of rainy days versus not-rainy days, in other words, what is what we call the **stationary distribution** $\vec{\pi}$ of probability among states?

$\vec{\pi} =$

In order for $\vec{\pi}$ to be well-defined, it must **not** depend on:

In fact no matter the initial distribution $\vec{v}_0$ we must have:

$$\lim_{n \rightarrow \infty} \vec{v}_0 P^n =$$

As a consequence: $\vec{\pi} P =$

A Markov chain is said to be **ergodic** if the **stationary distribution** $\vec{\pi}$ of the chain is well-defined, in which case:

The stationary distribution is also called the **stationary measure**. The vector $\vec{\pi}$ is a special left-eigenvector for the transition matrix **P** with eigenvalue $\lambda = 1$.

We will explain later why the chain of rainy versus not-rainy days is ergodic. First, let's assume it is, and then calculate the stationary distribution $\vec{\pi}$.